

Low-Power FPGA-Based Lunar Structure Matching Using Difference of Gaussian Filtering

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Abstract—Lunar missions need onboard image processing that runs fast, sips power, and fits hardware with tight resource limits. We present an FPGA image-filtering pipeline for lunar terrain matching built around the Difference of Gaussian (DoG) operator, a band-pass filter that picks out mid-frequency structure like crater rims and terrain edges. These features are prominent at high lunar latitudes, where low solar angles enhance surface contrast and favor gradient-based filtering. The system receives image data through a UART interface, runs two parallel separable Gaussian convolution stages with different smoothing kernels, and stores in on-chip block RAM (BRAM). The host system then applies normalized cross-correlation (NCC) to compare DoG output maps against stored reference maps. The architecture is evaluated using Chandrayaan-2 Terrain Mapping Camera (TMC) imagery from ten regions in the northern lunar highlands. Regions with a well-defined crater-rim structure achieve NCC match rates ranging from 73 to 99%. The design consumes an estimated 95 mW of on-chip power and occupies only 3.13% of the available LUTs on the target device. Overall, the proposed approach offers a low-power, resource-efficient solution for terrain matching onboard in future lunar missions.

Index Terms—FPGA, Difference of Gaussian, feature extraction, terrain-relative navigation, embedded vision

I. INTRODUCTION

Autonomous vision-based navigation is essential for planetary exploration tasks such as terrain-relative navigation (TRN), hazard detection and avoidance (HDA), and orbital change detection. Chandrayaan-3 has shown that onboard vision must work reliably under strict real-time and power constraints [1]. FPGAs suit this role, offering deterministic latency, reconfigurability, and better radiation tolerance than general-purpose processors [2], [3]. Lunar surface appearance varies with latitude: near-vertical equatorial sunlight minimises shadow length and contrast, while at higher latitudes low, grazing sunlight casts long shadows that sharply enhance crater-rim and terrain-boundary contrast. This makes band-pass filtering methods such as DoG well suited to polar and sub-polar terrain, where shadow edges carry clear structural information at scales a fixed hardware filter can capture. The high density of well-preserved impact craters at these latitudes [4] further enriches the available structural content. Full descriptor pipelines such as SIFT [5] and ORB [6] are robust and widely used, but their cost can be impractical on severely resource-constrained hardware. DoG, used here as a standalone filter rather than part of a full SIFT pipeline,

extracts the same mid-frequency structural signal at a fraction of the cost. For orbital imagery from consistent altitudes, where the main challenges are illumination variation, minor rotation, and sensor noise rather than large viewpoint change, the resulting structure-energy map can be matched directly via NCC without computing full descriptors. This work makes the following contributions:

- A fully implemented FPGA dual-kernel DoG pipeline on a Xilinx Artix-7 (Basys 3), using separable 5-tap binomial approximation kernels at two smoothing scales, a 2 Mbaud UART streaming interface, and runtime-configurable image dimensions.
- A hardware-software co-design evaluation framework running NCC-based template matching on a host PC, tested on 1000 augmented patches from 10 regions of Chandrayaan-2 TMC highland imagery, analyzing which terrain types the single-scale pipeline reliably handles.

II. RELATED WORK

A. Feature Matching Methods and Lunar Context

Classical feature descriptors trade invariance against computational cost, with direct implications for space hardware. SIFT detects keypoints as extrema in a DoG scale-space and computes a gradient-based descriptor robust to scale, rotation, and illumination change. ORB uses binary oriented FAST (Features from Accelerated Segment Test) keypoints and rotated BRIEF (Binary Robust Independent Elementary Features) descriptors, running substantially faster than SIFT at the cost of some scale-change robustness. For orbital lunar imagery at consistent altitudes, where illumination variation and minor rotation dominate over extreme viewpoint change, the full machinery of SIFT or ORB is unnecessary. This work instead uses DoG as a standalone filtering stage, matching the resulting structure-energy map directly via NCC. A recent alternative abandons classical descriptors for learned ones: MARs [7] trains a deep metric-learning network for lunar/Martian crater-landmark re-identification, targeting recognition accuracy under extreme inter-class similarity rather than the minimal power and area emphasised here (Table V).

B. FPGA Feature Extraction

Gupta *et al.* [8] survey FPGA-based feature extraction architectures and trade-offs in throughput, resource usage, and

precision. Bukhari and Iqbal [9] propose a dedicated hardware architecture for DoG calculation, directly relevant to this work. SIFT-specific FPGA implementations include Yao *et al.* [10] (hardware-efficient keypoint extraction), Reddy *et al.* [11] (full SIFT detector with optimised convolution), and parallel architectures exploiting spatial concurrency [12]. Mukherjee *et al.* [13] propose a fast fixed-point 2-D Gaussian filter architecture, a relevant baseline for the convolution stages underlying DoG. Jamali *et al.* [14] demonstrate energy-efficient parameterised 2-D separable convolution on FPGA with significant resource savings over direct 2-D implementations, borne out in our design. Two recent designs contextualise the present pipeline’s 95 mW, 3.13%-LUT footprint (Table III). Wan *et al.* [15] implement a full ORB frontend on a Zynq UltraScale+ XCZU9EG for quad-camera odometry, reporting 1.63 W at 51% LUT for 640×480 images on a substantially larger device. Chen *et al.* [16] implement a JPEG XS displacement-vector search block on the same XC7A35T used here, reporting 277 mW at 100 MHz. This is a different domain (video coding), but it offers a same-platform reference point.

C. Vision for Space Applications

Vision-based navigation is well established in space missions. Chandrayaan-3 employed a vision-based HDA system for landing site selection [1], showing onboard vision is operationally viable under a lander’s severe constraints. Chandrayaan-2-based feature matching is studied in [17]. Pérez *et al.* [2] present a run-time reconfigurable MPSoC-based on-board processor for vision-based space navigation, and Thomas *et al.* [3] adapt robotics vision algorithms for space-rated FPGAs. This work complements that literature with a low-power, modular FPGA band-pass pipeline on real Chandrayaan-2 TMC highland imagery, analysing which terrain types and latitude-dependent illumination benefit from single-scale DoG filtering.

III. SYSTEM ARCHITECTURE

The system follows a hardware-software co-design philosophy: the FPGA performs all convolution at deterministic latency, while the host PC manages dataset operations, augmentation, and NCC matching (Fig. 1).

A. FPGA Pipeline

The FPGA pipeline runs on a Xilinx Artix-7 XC7A35T (Basys 3) at 100 MHz in three stages.

1) *Input Interface*: A UART receiver at 2 Mbaud (50 clocks/bit at 100 MHz) streams 8-bit grayscale pixels from the host. A 0x55 sync byte marks frame boundaries. Pixels are stored sequentially into a 16 KB dual-port BRAM (input buffer), supporting up to 128×128 pixels.

2) *DoG Core*: Two independent separable convolution pipelines operate in parallel on the same input BRAM, sharing a read address bus for cycle-accurate sync. Each applies a separable 5-tap kernel using:

- A horizontal stage using a shift-register tap chain.

- A vertical stage using four 128-entry line buffers to maintain a five-row sliding window.

The two kernels are 5-tap binomial approximations to Gaussian smoothing at two different smoothing scales:

$$\mathbf{k}_1 = \frac{1}{16}[1, 4, 6, 4, 1] \quad (1)$$

$$\mathbf{k}_2 = \frac{1}{8}[1, 2, 2, 2, 1] \quad (2)$$

These are integer-coefficient binomial approximations, not analytically defined Gaussian kernels with precise σ . $\mathbf{k}_1$ imposes broader smoothing than $\mathbf{k}_2$. The structure-energy response is their difference:

$$\text{DoG}(x, y) = B_1(x, y) - B_2(x, y) \quad (3)$$

where B_1 and B_2 are the outputs of the respective blur stages. The result is a signed 9-bit value with data-dependent dynamic range. Unlike SIFT, which searches for signed extrema in a scale-space pyramid, this design uses the *magnitude* of $\text{DoG}(x, y)$ as a spatial structure-energy map matched via NCC. The result is a hardware band-pass filter whose output encodes local edge and structural energy, capturing shadow-boundary signal directly at high-latitude crater rims. Each pipeline uses a three-state FSM (IDLE → READ → FLUSH) with a latency compensation counter to discard the initial pipeline fill cycles. Image width and height are passed as runtime 8-bit parameters, allowing flexible patch sizes without re-synthesis.

3) *Mapping and Output Interface*: The signed DoG pixel is mapped to 8-bit unsigned format before transmission:

$$p_{\text{out}} = 128 + \left\lfloor \frac{\text{DoG}}{2} \right\rfloor \quad (4)$$

This centres the output at 128 (zero response), preserving sign information for the host. In Verilog: $128 + (\text{dog} \ggg 1)$, where $\ggg$ is a sign-preserving arithmetic right shift on the signed value. Mapped pixels are written to a second 16 KB output BRAM. The top-level FSM then transmits a 0x55 sync byte followed by the 16,384-byte output frame via UART TX. The top-level controller manages five states:

- 1) **S_WAIT_SYNC**: Idle. Waits for 0x55 from the host.
- 2) **S_RX_FRAME**: Receives 16,384 pixels into input BRAM.
- 3) **S_PROCESS**: Asserts the start signal to the DoG core. Waits for the done signal.
- 4) **S_TX_SYNC**: Sends 0x55 back to signal readiness.
- 5) **S_TX_FRAME**: Streams 16,384 output pixels to the host.

B. Host Processing Pipeline

The host (PySerial/OpenCV) handles dataset management, FPGA communication, and NCC matching. For each frame:

- 1) Resize to 128×128. Replace sync byte (0x55) with 0x54 to prevent false sync.
- 2) Transmit to FPGA. Receive DoG output.
- 3) Remove the 128-offset bias to recover the signed response. Take the absolute value for a structure-energy

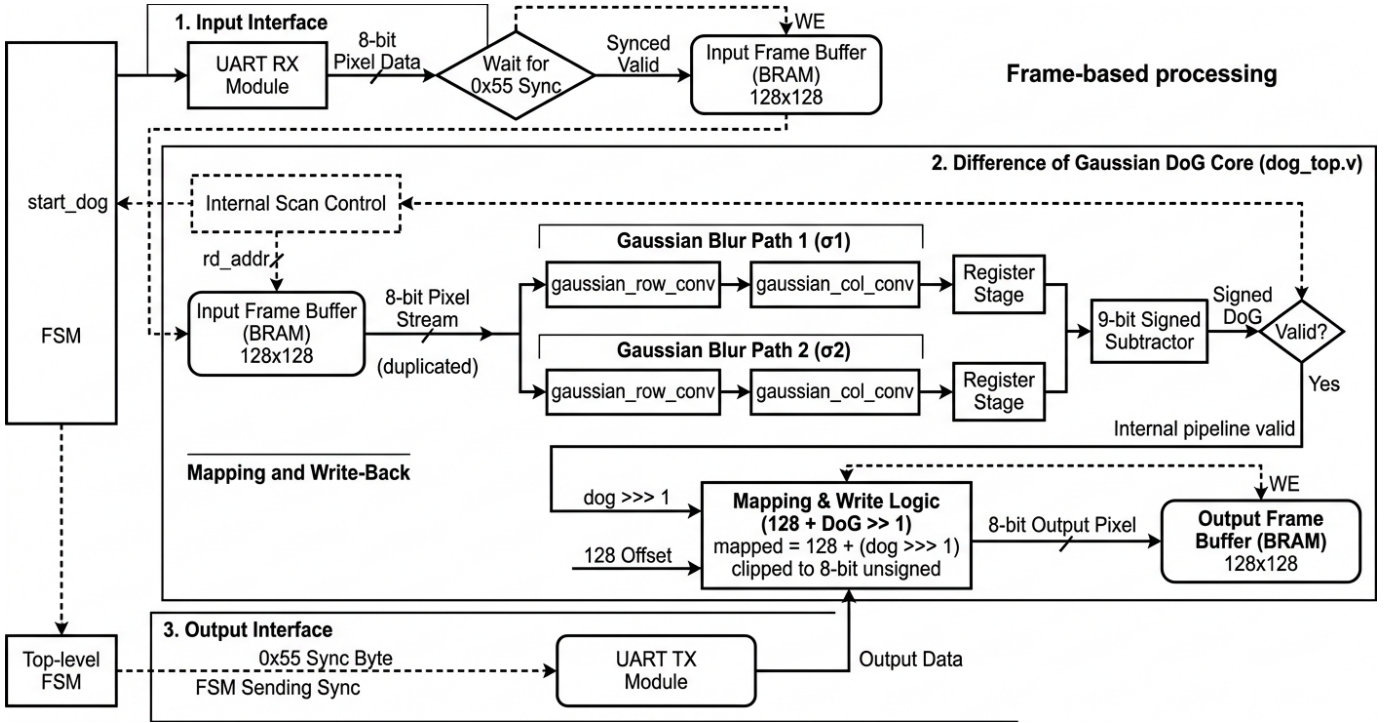


Fig. 1: FPGA DoG processing pipeline. Frames are streamed in via UART RX, buffered in input BRAM, processed through two parallel separable convolution paths, and the mapped DoG output is transmitted via UART TX. The mapping operation implements $p_{out} = 128 + \lfloor \text{DoG}/2 \rfloor$ using an arithmetic right shift on the signed DoG value.

magnitude map. Apply Gaussian smoothing and min-max normalisation to $[0, 1]$. Match against the stored template energy map via NCC.

4) Threshold 0.5 classifies match / no-match.

IV. METHODOLOGY

A. Lunar Terrain Context and Image Source

The dataset is a Chandrayaan-2 TMC nadir-channel calibrated browse image of the northern lunar highlands, accessed via the ISRO ISSDC Map Browse portal [18]. The strip covers the Aristoteles crater neighbourhood, approximately 41°N – 61°N , 26°E – 28°E , at ~ 18 m/pixel resolution.

B. Patch Extraction and Geo-referencing

Ten 128×128 patches are extracted using a texture score combining pixel variance and mean absolute Sobel gradient magnitude, with ≥ 150 px spatial separation between centres. Geo-referencing uses the accompanying geometry CSV to map pixel coordinates to lunar latitude/longitude. The patches span $\sim 41^\circ\text{N}$ to $\sim 60^\circ\text{N}$, covering smoother plains to denser, rugged crater fields, enabling an assessment of latitude-dependent terrain characteristics, as shown in Fig. 2 and Table I.

C. Augmentation Protocol

Each patch is augmented with small rotations (up to 10°), scale variations ($\pm 5\%$), Gaussian blur, and additive noise to emulate different imaging conditions. Per region, this produces one Gaussian-blurred, min-max normalised template

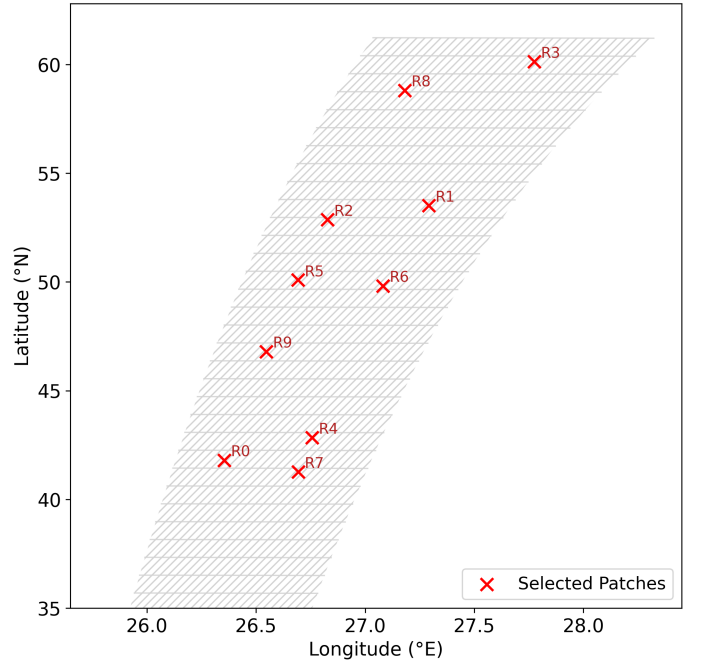


Fig. 2: Geographic distribution of the 10 selected 128×128 patches (R0–R9) within the TMC strip footprint ($\sim 41^\circ\text{N}$ to 60°N , 26°E to 28°E).

TABLE I: Dataset Regions – Coordinates and Visual Characterization

R	Lat (°N)	Lon (°E)	Visual character
0	41.80	26.36	Flat; visually featureless at 128 px
1	53.51	27.29	Strong crater rim, high contrast
2	52.86	26.83	Dense crater structure
3	60.12	27.78	Rugged multi-directional edges
4	42.84	26.76	Moderate structured relief
5	50.09	26.69	Low visual contrast
6	49.82	27.08	Low visual contrast
7	41.27	26.69	Visually smooth
8	58.81	27.18	Sparse, wide crater floors
9	46.80	26.55	Intermediate texture

patch, plus 25 transformed frames with varying rotation/scale. Stochastic Gaussian blur and noise ($\sigma = 3$) are applied with probability 0.3 for realism. Each of the 25 frames per region is evaluated under four host-side augmentation variants: (1) **Original**: frame as generated. (2) **Noise**: Gaussian noise ($\sigma = 10$) added. (3) **Rotated**: additional 8° rotation. (4) **Brightness**: scaled by 1.2 with +10 offset. This yields $10 \times 25 \times 4 = 1000$ total FPGA inference calls.

V. RESULTS AND DISCUSSION

The design is implemented on a Xilinx Artix-7 XC7A35TCPG236-1 (commercial grade, typical corner) in Vivado 2016.1, with timing closure confirmed at WNS = +2.893 ns, zero failing endpoints across 2260 paths (Table II). 3.13% LUT utilization confirms the Artix-7

TABLE II: Post-Implementation Resource Utilization (XC7A35T)

Resource	Used	Avail.	%	Note
LUT (Logic)	650	20,800	3.13	
LUT (Dist. RAM)	96	9,600	1.00	Line buffers
Flip-Flops	540	41,600	1.30	
Block RAM	9	50	18.00	2× frame BRAM
I/O	20	106	18.87	UART + LED
BUFG	1	32	3.13	System clock

35T’s suitability, with headroom for additional processing stages. Separable convolution [14] is the primary reason for this low footprint: it reduces per-pixel MAC count from $O(N^2)$ to $O(N)$, i.e. 10 multiplications instead of 25 per pixel per stage for $N = 5$. This is central to the design’s claim as a lightweight complement to SIFT/ORB. The full dual-kernel DoG pipeline fits in roughly the area a single SIFT Gaussian blur stage would need. The eight 128-entry line buffers in the column-convolution stages are inferred as distributed RAM (96 LUTRAMs), avoiding extra BRAM. The 18% BRAM utilization is the two 16 KB frame buffers (input/output). Post-implementation power estimation (default switching activity, medium confidence) puts total on-chip power at 95 mW, split across Vccint (9.8 mW, 1.0 V), Vccaux (22.7 mW, 1.8 V), Vcco33 (3.3 mW, 3.3 V),

Vccbram (0.4 mW, 1.0 V), and Vccadc (36.0 mW, 1.8 V), with estimated junction temperature 25.5°C (thermal margin $59.5^\circ\text{C} / 11.8 \text{ W}$, $\theta_{JA} = 5.0^\circ\text{C/W}$). These are estimates, not measurements, and should be validated by hardware measurement before flight use. Table III contextualises this footprint against two recent FPGA vision/pattern-matching designs (Section II-B). These span two device generations and domains, so the comparison is contextual rather than direct. On identical XC7A35T silicon, the JPEG XS core’s 277 mW

TABLE III: Hardware Comparison With FPGA Based Designs

Design	Platform	Power	LUT %
Ours (dual DoG)	XC7A35T (Artix-7)	95 mW	3.13%
JPEG XS DV search [16]	XC7A35T (Artix-7)	277 mW	⁻¹
Quad-camera ORB [15]	XCZU9EG (Zynq US+)	1630 mW	51%

($\sim 2.9\times$ ours) reflects its larger candidate-search engine (11.63K logic LUTs vs. 650 here) against this work’s fixed, search-free convolution. The quad-camera ORB frontend’s 1.63 W/51% LUT on a substantially larger UltraScale+ device reflects added detection, descriptor, and matching cost. Both corroborate Table V: no descriptor or search stage keeps power and area an order of magnitude below full-pipeline designs, at the cost of narrower applicability (Section VI-C). At 2 Mbaud, transferring one 128×128 frame takes ~ 82 ms per direction (~ 164 ms round-trip), consistent with the measured 0.187 s/variant average (difference from host-side serialisation and NCC). The convolution pipeline itself processes all 16,384 pixels at 100 MHz in a few hundred microseconds, making UART the dominant latency bottleneck.

A. Per-Region Analysis

1000 FPGA round-trips complete in 187 s (18.7 s/region, 100 variants each). At threshold 0.5, the overall NCC match rate is 35.7%, and scores split into two clear populations: non-matching variants sit in the 0.1–0.3 range, matching ones in 0.5–0.7 (Table IV). This split tracks how well each

TABLE IV: Per-Region NCC Score Statistics (100 Variants Each)

R	Mean	Min	Max	Match%
0	0.031	−0.004	0.100	0%
1	0.564	0.477	0.705	89%
2	0.531	0.413	0.655	73%
3	0.654	0.488	0.771	99%
4	0.634	0.325	0.782	96%
5	0.279	0.128	0.476	0%
6	0.232	0.093	0.460	0%
7	0.185	0.074	0.430	0%
8	0.261	0.061	0.493	0%
9	0.196	0.086	0.383	0%
All	0.357	−0.004	0.782	35.7%

region’s spatial structure aligns with the kernel’s passband. **Regions 1–4** (near Aristoteles M/N craters, $\sim 42\text{--}60^\circ\text{N}$) give the dataset’s best scores (Table IV). Their compact crater rims and rugged, multi-directional edges catch oblique light as sharp, repeatable shadows, so the structure-energy maps come out consistent frame to frame. Region 3 tops the list. A couple of small dips have clear causes: Region 2 loses some ground to its rotated variant, and Region 4 has one rotated-frame outlier (NCC = 0.325), probably from the sinusoidal transform clipping the patch edge. Neither dents the region’s overall match rate.

Regions 0 and 5–9, by contrast, stay below threshold throughout. Region 0 (41.8°N) is the more telling case. What looks like texture at full resolution turns out to be sub-pixel noise that disappears on downsampling, leaving a nearly flat DoG response. This is consistent with the shorter shadows and gentler relief typical of this latitude. Regions 5–9 fail for a different reason: their wide crater floors, gradual albedo gradients, and smooth plains carry structure at scales too large for the 128-pixel patch or outside the DoG passband altogether, so edge energy exists but is too spread out to match on. Neither failure mode points to a hardware limitation. Both come down to a scale or contrast mismatch with the pipeline’s parameters. Fig. 3 shows representative patches from the three highest-performing regions.

B. Band-Pass Filtering, Illumination, and Scale

The results back up the underlying physical reasoning: oblique light at higher latitudes sharpens crater-rim shadow contrast into mid-frequency edge energy that the DoG band-pass picks up reliably, while at lower latitudes and over smooth terrain the energy map flattens out regardless of how visually rich the scene looks, and NCC fails accordingly. The 10 patches were chosen to span diverse highland terrain rather than to favour easy cases, precisely so we could pin down which terrain types and scales suit this pipeline versus where a heavier method like SIFT or ORB would be needed.

C. Architectural Tradeoffs, Complexity, and Threshold Sensitivity

Streaming line buffers in the column-convolution stage save us a second large BRAM, but at a cost: processing has to run sequentially and single-pass, since restarting mid-frame means flushing all four line buffers first. UART at 2 Mbaud is the real latency bottleneck, dominating the round trip at ≈ 164 ms. A faster link such as SPI, LVDS, or MIPI CSI-2 would cut that transfer time by 1–2 orders of magnitude and open the door to video-rate operation. On thermal margin, the estimated 95 mW power draw and 25.5°C junction temperature look comfortable, though that still needs confirming on real hardware. Table V places the pipeline next to established FPGA and learned-descriptor methods, comparing per-pixel compute and pipeline memory as functions of kernel half-width N , image dimensions $W \times H$, scale levels S , and keypoints K . The line-buffer architecture keeps pipeline memory down to $O(NW)$, using four 128-entry line buffers per

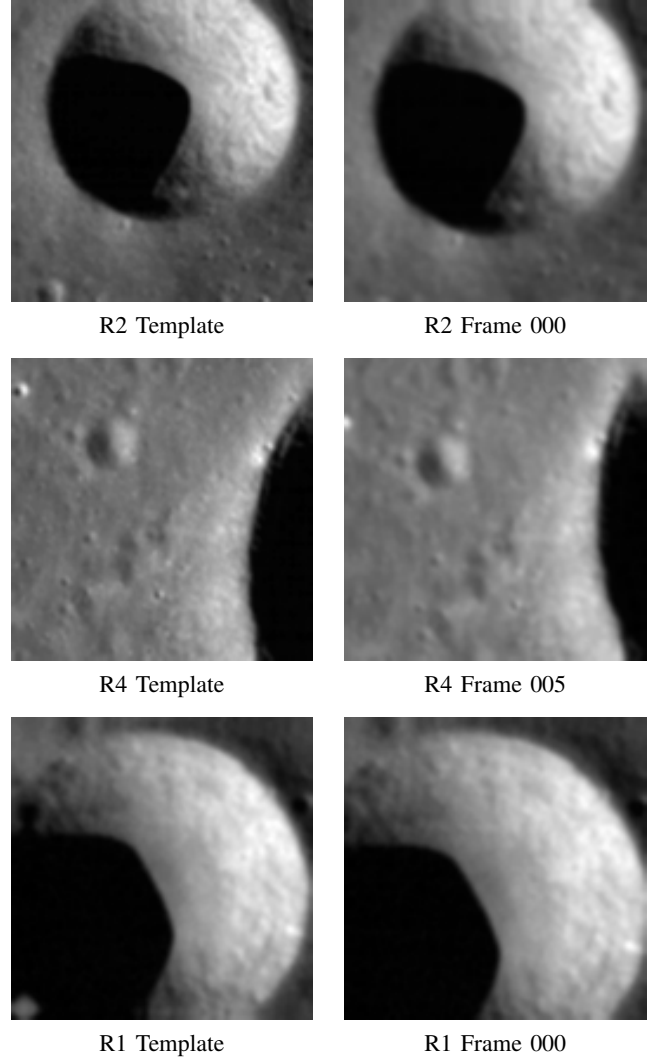


Fig. 3: Representative 128×128 patches from the three highest-scoring regions. Each row shows a template (left) and a corresponding frame (right). Crater rims, shadow boundaries, and multi-directional edges provide the spatial structure preserved by the DoG band-pass filter. These features are characteristic of higher-latitude lunar highland terrain under oblique solar illumination.

TABLE V: Algorithmic Complexity Comparison

Method	Per-pixel compute	Pipeline memory	Desc.
2-D direct conv.	$O(N^2)$	$O(WH)$	No
SIFT [10], [12]	$O(SN^2) + O(K \cdot 128)$	$O(SWH)$	Yes
ORB [6]	$O(P) + O(K \cdot 32B)$	$O(WH)$	Yes
Sep. conv. [14]	$O(N)$	$O(NW)$	No
MARs [7]	CNN forward pass ($\sim 44\text{M}$ params, ResNeXt-101)	Model weights + activations	Yes
Ours (dual DoG)	$2O(N)$	$2O(NW)$	No

pipeline. This is well below the $O(WH)$ full-frame store a direct 2-D convolution would need. Running two pipelines side by side just doubles the constant factor without changing that order. Full descriptor pipelines carry extra overhead on top: $O(S \cdot WH)$ for scale-pyramid storage and $O(K \cdot 128)$ per-keypoint descriptor memory, needed to support six-degree-of-freedom relocalisation across arbitrary viewpoints. Learned approaches like MARs [7] take a different route entirely, running a full network forward pass (622–12076 ms on space-representative hardware). This is a large latency cost in exchange for better discrimination between craters that look nearly identical. Leaving out descriptor computation here is a deliberate choice: at consistent orbital altitudes, where illumination change and minor rotation are the main challenges for TRN, the structure-energy map alone is enough for NCC matching. The 0.5 NCC threshold is on the conservative side for Regions 1–2, where near-miss variants still score 0.47–0.49 while every failing region averages below 0.28. Dropping the threshold to 0.45 would pick up more recall without giving up much precision.

VI. CONCLUSION

We built and evaluated a real-time FPGA dual-kernel DoG band-pass filter for structure-energy extraction from lunar imagery. On a Xilinx Artix-7 XC7A35TCPG236-1 (Basys 3), the design closes timing at +2.893 ns WNS, draws an estimated 95 mW, and uses just 3.13% of available LUTs. This is well within reach of a space-grade embedded budget and far lighter than a full SIFT or ORB implementation. Testing against 10 spatially distributed Chandrayaan-2 TMC highland patches shows a clear latitude split: mid-to-high-latitude, crater-rich terrain with strong shadow contrast matches reliably (73–99% NCC), whereas smooth equatorward terrain does not. That split points to single-scale DoG filtering, rather than the hardware itself, as the current bound on terrain coverage. A multi-scale DoG hierarchy would broaden the set of matchable terrain types. Orientation-aware descriptors could help on low-texture regions. The present prototype also depends on a host laptop for image transfer and NCC matching over UART, which adds communication overhead and host-side computation. Replacing UART with a faster link such as SPI, LVDS, or MIPI CSI-2 would remove this bottleneck and enable video-rate operation. Interfacing the imaging sensor directly with the FPGA, together with a lightweight neural-network-based similarity engine in place of host-side NCC matching, would move the system toward a fully embedded, real-time design. On-hardware power measurement is still needed to confirm the present estimates, and testing on polar TMC strips beyond 70°N would show whether more grazing illumination helps DoG matching further.

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